(i)

$$\begin{aligned}\frac{d}{dx}\left(\frac{P(x)}{Q(x)}e^x\right) &= \left(\frac{P(x)}{Q(x)}\right)' e^x + \frac{P(x)}{Q(x)}e^x \\ &= \frac{P'(x)Q(x) - P(x)Q'(x)}{Q(x)^2}e^x + \frac{P(x)Q(x)}{Q(x)^2}e^x \\ &= \frac{e^x\left(Q(x)(P(x)+P'(x)) - P(x)Q'(x)\right)}{Q(x)^2} \\ &= \frac{x^3-2}{(x+1)^2}e^x\end{aligned}$$

$$\therefore \big[Q(x)(P(x)+P'(x)) - P(x)Q'(x)\big](x+1)^2 = (x^3-2)Q(x)^2$$

When $x=-1 \implies 0 = -3Q(-1)^2 \implies Q(-1)=0$

$$\therefore \boxed{x+1 \text{ is a factor of } Q(x)}$$

Let $\deg(P(x)) = p,\ \deg(Q(x)) = q \implies \deg(P(x)+P'(x)) = p,\ \deg\big(Q(x)(P(x)+P'(x))\big) = p+q,\ \deg\big(P(x)Q'(x)\big) = p+q-1$

$$\deg\left(Q(x)(P(x)+P'(x)) - P(x)Q'(x)\right) = p+q$$

$$\therefore \deg\left(\big[Q(x)(P(x)+P'(x)) - P(x)Q'(x)\big](x+1)^2\right) = p+q+2$$

$$\deg\left((x^3-2)Q(x)^2\right) = 2q+3$$

$$p+q+2 = 2q+3 \implies p = q+1$$

$$\therefore \boxed{\deg(P(x)) = \deg(Q(x)) + 1}$$

When $Q(x) = x+1,\ Q'(x) = 1$

$$\begin{aligned}\big[Q(x)(P(x)+P'(x)) - P(x)Q'(x)\big](x+1)^2 &= (x^3-2)Q(x)^2 \\ \big[(x+1)(P(x)+P'(x)) - P(x)\big](x+1)^2 &= (x^3-2)(x+1)^2 \\ (x+1)(P(x)+P'(x)) - P(x) &= x^3-2 \\ xP(x) + (x+1)P'(x) &= x^3-2\end{aligned}$$

Let $P(x) = ax^2+bx+c$

$$\begin{aligned}xP(x) + (x+1)P'(x) &= x(ax^2+bx+c) + (x+1)(2ax+b) \\ &= ax^3 + (b+2a)x^2 + (c+2a+b)x + b\end{aligned}$$

$$\begin{cases} a = 1 \\ b + 2a = 0 \\ c + 2a + b = 0 \\ b = -2 \end{cases} \implies \begin{cases} a = 1 \\ b = -2 \\ c = 0 \end{cases}$$

$$\therefore \boxed{P(x) = x^2 - 2x}$$

(ii)
Assume such $P(x), Q(x)$ exist and $\gcd(P, Q) = 1$

$$\frac{d}{dx}\left(\frac{P(x)e^x}{Q(x)}\right) = \frac{1}{x+1}e^x$$
$$\frac{e^x\left(Q(x)(P(x)+P'(x)) - P(x)Q'(x)\right)}{Q(x)^2} = \frac{1}{x+1}e^x$$

$$\therefore \Big[Q(x)(P(x)+P'(x)) - P(x)Q'(x)\Big](x+1) = Q(x)^2$$

When $x = -1 \implies 0 = Q(-1)^2 \implies Q(-1) = 0$
Let $Q(x) = (x+1)R(x)$

$$Q'(x) = R(x) + (x+1)R'(x)$$

$$(x+1)R(x)(P(x)+P'(x)) - P(x)(R(x)+(x+1)R'(x)) = (x+1)R(x)^2$$

When $x = -1 \implies -P(-1)R(-1) = 0$
$\because \gcd(P, Q) = 1$ and $x + 1 \mid Q(x) \quad \therefore R(-1) = 0 \implies (x+1)^2 \mid Q(x)$
Let $Q(x) = (x+1)^n S(x), \qquad n \geq 2, \qquad S(-1) \neq 0$

$$Q'(x) = n(x+1)^{n-1}S(x) + (x+1)^n S'(x)$$

$$\Big[Q(x)(P(x)+P'(x)) - P(x)Q'(x)\Big](x+1) = Q(x)^2$$

$$(x+1)^n\Big[(x+1)S(x)(P(x)+P'(x)) - P(x)(nS(x)+(x+1)S'(x))\Big] = (x+1)^{2n}S(x)^2$$

$$(x+1)S(x)(P(x)+P'(x)) - P(x)(nS(x)+(x+1)S'(x)) = (x+1)^n S(x)^2$$

When $x = -1 \implies -nP(-1)S(-1) = 0$

$$\because n \neq 0, \qquad P(-1) \neq 0, \qquad S(-1) \neq 0$$

Contradiction

$\therefore$ $\boxed{\text{No such polynomials } P(x), Q(x)}$

This solution was submitted by **Jon**

Published and refined by **Jon**

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